Compact CNNs for Chest X-ray Screening: Accuracy, Explainability, and Containerised Deployment

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Abstract

Lightweight convolutional networks are widely promoted for medical-image triage on resource-constrained hardware, on the assumption that smaller models translate into faster, cheaper deployment. We test this in a pilot study on a three-class chest X-ray task (Normal, Pneumonia, Tuberculosis) assembled from four public sources, with patient-grouped splits verified disjoint by path, content hash, patient group and pairwise pixel correlation. A custom 6,627-parameter CNN (*Victorio*) is compared against MobileNetV2, EfficientNet-B0 and ResNet-50 under two initialisation regimes on a held-out test set of 480 images. The three backbones are statistically indistinguishable at 97.3–98.3% accuracy across a tenfold parameter range, while Victorio reaches 84.2%, a 14 pp gap far outside the $\pm$1.6 pp resolution of the test set; pretraining helps every backbone by 1.7 to 6.6 pp. A Grad-CAM versus LIME Agreement Index, computed from raw attribution arrays and stable under three mask thresholds, ranges from 0.13 to 0.24 and varies with architecture and regime rather than being invariant to capacity. A source classifier recovers the acquisition dataset within a fixed diagnosis at 0.98 balanced accuracy, confirming a source confound; performance nonetheless holds on a single-source subset. A deployment benchmark (5,040 requests) exposes an architecture-dependent CPU/GPU crossover: GPU serves ResNet-50 at 2.1× the CPU median, ties on MobileNetV2, and loses to CPU by 2.5× on the compact network.

Keywords: Chest X-ray, CNN, Transfer learning, Explainability, Kubernetes deployment.

1 Introduction

Chest radiography remains the most common front-line imaging modality for respiratory disease, and convolutional neural networks (CNNs) routinely match or exceed expert performance on curated benchmarks [Rajpurkar et al., 2017]. Lightweight architectures such as MobileNetV2 [Sandler et al., 2018] and EfficientNet [Tan and Le, 2019] are increasingly favoured for clinical deployment, with the implicit claim that fewer parameters translate into faster, cheaper inference. In practice, end-to-end response time in a containerised service depends on preprocessing, HTTP, orchestration and model compute combined; the parameter-count advantage of a minimal network can be absorbed by the surrounding stack.

This paper is a pilot study addressing two hypotheses: **(H1)** that an ultra-compact custom CNN matches transfer-learning baselines on a balanced three-class CXR task; and **(H2)** that the CPU/GPU latency advantage is architecture-dependent, with GPU acceleration amortising its per-request launch overhead on heavier backbones but losing that advantage on compact models. We make four contributions:

- A three-class CXR corpus from four public sources with patient-grouped splits, verified disjoint at four levels including pairwise pixel correlation over all 5.17 M cross-split pairs.
- A capacity comparison of a 6,627-parameter custom CNN against backbones spanning 2.2 M to 23.5 M parameters, with confidence intervals so the resolution limit of the test set is explicit.
- A source-confound analysis: a within-class pixel signature test, a source-discriminating classifier, and an evaluation on a single-source subset in which the confound cannot operate.

- A latency characterisation of the deployed service on a local Kubernetes cluster (5,040 requests, CPU and GPU pods).

2 Related Work

CNNs for chest radiography. Modern CXR pipelines almost universally rely on backbones pretrained on ImageNet, with reported AUROC values approaching radiologist performance for pneumonia and tuberculosis screening [Rajpurkar et al., 2017; Irvin et al., 2019; Liu et al., 2020]. Whether natural-image pretraining is the right prior for radiology is contested: RadImageNet [Mei et al., 2022] shows that pretraining on radiological images can outperform ImageNet initialisation on medical tasks, and comparisons of compact backbones on multi-label CXR classification [Ucan et al., 2025] report similar accuracy across the EfficientNet family, consistent with the capacity saturation we observe. CheXpert [Irvin et al., 2019] and TBX11K [Liu et al., 2020] are typically studied in isolation; cross-dataset fusion introduces source-class confounds [Zech et al., 2018] that must be measured rather than asserted away. Other work on medical-imaging CNN training addresses a different configuration axis: Parise and Mac Namee evaluate representation, model capacity and query strategy jointly, optimising labelling efficiency rather than the deployment characteristics studied here [Parise and Mac Namee, 2023].

Lightweight architectures and serving. Inverted residuals [Sandler et al., 2018] and compound scaling [Tan and Le, 2019] reduce parameter count by an order of magnitude while preserving accuracy on natural images, but reported improvements usually refer to model FLOPs rather than end-to-end response times. Serving systems such as Clockwork [Gujarati et al., 2020] target predictable per-request latency under containerised deployment, without decomposing it by architecture and backend at the granularity relevant to single-image medical inference.

Explainability. Grad-CAM [Selvaraju et al., 2017] and LIME [Ribeiro et al., 2016] are the two most common post-hoc attribution methods in radiology research. Quantitative comparisons between them are rare, and Adebayo et al. [2018] show that saliency methods can produce plausible maps that are insensitive to the model being explained, so any agreement metric requires a sanity check before it can be interpreted.

3 Materials and Methods

3.1 Dataset construction

The corpus draws on *four* public sources. The TBX11K release [Liu et al., 2020] bundles three further tuberculosis datasets under `imgs/extra/`: Montgomery County and Shenzhen [Jaeger et al., 2014], and DA and DB [Chauhan et al., 2014]. Table 1 gives the composition. Pneumonia is drawn from CheXpert-Small [Irvin et al., 2019] frontal views. The corpus is balanced at 1,072 images per class, 3,216 in total, split 2,255 / 481 / 480 into training, validation and test, with the test set balanced at 160 per class.

Class	TBX11K	Montgomery/Shenzhen	DA/DB	CheXpert
Normal	985	57	30	—
Tuberculosis	800	198	74	—
Pneumonia	—	—	—	1,072

Table 1: Corpus composition by class and acquisition source, 1,072 images per class. Pneumonia originates from a single source, which is the basis of the confound analysed in Section 5.3.

Class definition. The tuberculosis class comprises radiographs carrying any TB annotation. TBX11K distinguishes active from obsolete (healed, inactive) disease: of the 1,072 TB images, 58.8% are annotated active, 13.0% obsolete, 2.8% both, and 25.5% come from sources that record TB positivity without activity sub-classification. Reported tuberculosis recall is therefore sensitivity to TB-associated radiographic findings, not to active infection. Figure 1 shows representative radiographs from each class.

Split integrity. CheXpert is split by patient (901 patients, 1.19 images each); the remaining sources contribute one image per subject. Splits were verified disjoint by file path, SHA-256 content hash, patient group, and

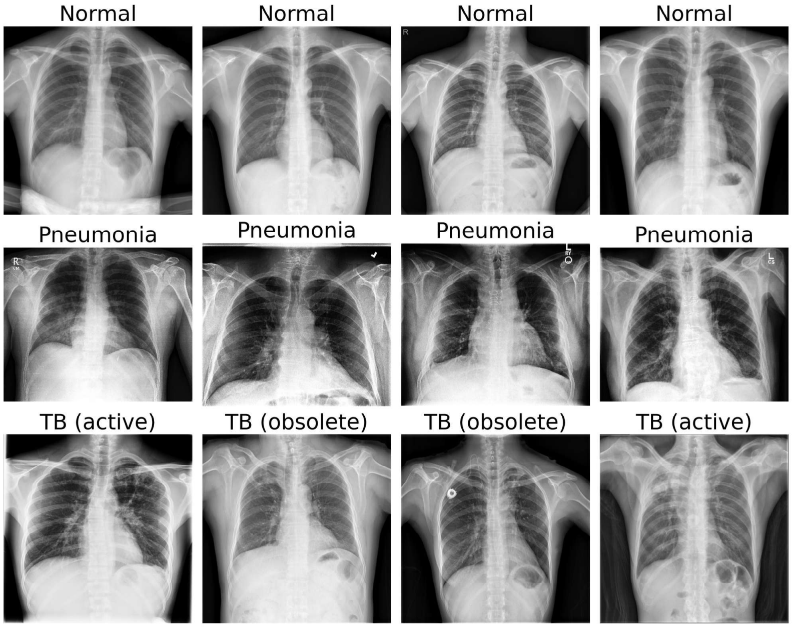

Figure 1: Representative radiographs from the test split, four per class. The tuberculosis panels are annotated with their activity label, illustrating that the class contains both active and healed disease.

pairwise Pearson correlation of z-scored 64×64 images over all 5,169,720 cross-split pairs. The maximum cross-split similarity is 0.9722, against approximately 0.9999 for a re-encoded duplicate, and the pair concerned is retained and disclosed rather than removed, so the test set remains exactly reproducible from the build script.

3.2 Architectures

MobileNetV2 [Sandler et al., 2018] (2,227,715 parameters), EfficientNet-B0 [Tan and Le, 2019] (4,011,391) and ResNet-50 [He et al., 2016] (23,514,179) were used with the final classification head replaced by a 3-way linear layer; counts are for the 3-class configuration rather than the 1000-class ImageNet original. These three are referred to as the backbones throughout; Victorio is treated separately. **Victorio** is a custom minimal CNN of 6,627 parameters: two 3×3 convolutional blocks ($3 \rightarrow 16 \rightarrow 32$ channels), each followed by ReLU and 2×2 max-pooling, adaptive average pooling to 4×4, and a 3-way linear head, with no batch normalisation or dropout. The design intentionally explores the lower bound of CNN capacity for this task.

3.3 Training regimes

Each architecture was trained under two regimes. The **ImageNet** regime initialises the three backbones with ImageNet weights and uses ImageNet normalisation. The **CXR** regime initialises from random weights and applies fixed dataset-specific normalisation (per-channel mean 0.534, standard deviation 0.266), applied identically to training, validation and test. Images are single-channel grayscale replicated across three RGB channels.

The two quantities a regime fixes, initialisation and normalisation, are logically separate and coincide only where pretrained weights exist. No ImageNet weights exist for Victorio, so both of its runs begin from random initialisation and differ in normalisation alone. Its regime comparison therefore answers a different question from that of the three backbones, and its rows in Table 2 must not be read as evidence about pretraining. A run manifest recording the initialisation and normalisation actually used is written alongside every checkpoint.

4 Experimental Setup

4.1 Training protocol

All models were trained with identical hyperparameters: input 256×256, batch size 128, 60 epochs, AdamW ($lr = 10^{-4}$, weight decay 10^{-4}), cosine schedule with 3-epoch warm-up, label smoothing 0.05, mixed precision, early stopping (patience 8). Augmentation was random horizontal flip only. Resizing used aspectratio-preserving letterbox padding to avoid clipping costophrenic angles. All trainable parameters were updated end-to-end. Best-validation checkpoints were retained. Using one hyperparameter set across architectures spanning four orders of magnitude in capacity is a deliberate control for fairness, but it disadvantages models whose optimum lies elsewhere; this is revisited in Section 6.

4.2 Evaluation and explainability

Accuracy, per-class precision, recall and F1, and one-vs-rest macro AUROC were computed on the held-out test set ($n = 480$). Every proportion carries a Wilson score interval, preferred over the normal approximation because the values lie close to 1.0 with small n. The interval half-width is approximately ± 1.6 pp overall and ± 2.9 pp per class at the observed accuracies, so differences below roughly 3 pp are not resolvable.

Grad-CAM [Selvaraju et al., 2017] maps were taken as the raw normalised activation array of the last activated convolutional block; the mask retains the highest-activating 20% of pixels by rank. LIME [Ribeiro et al., 2016] explanations used 1,000 perturbations per image, and the mask is the union of superpixels with positive fitted weight, read from the explanation object. Neither mask is derived from a rendered figure. The Agreement Index is the intersection over union of the two masks, computed over 20 test images per class for each of the eight model–regime pairs, retaining those the model classifies correctly, and repeated at mask thresholds of 10%, 20% and 30% to test threshold dependence.

4.3 Deployment protocol

Each checkpoint was packaged in a containerised FastAPI service with a single `/predict` POST endpoint, all eight served from one pod via a model query parameter. CPU (`python:3.10-slim`) and GPU (`nvidia/cuda:12.8.0-runtime-ubuntu22.04`) images were built with Podman and deployed as separate Pods on a local Minikube cluster (Docker driver, 6 vCPU, 16 GB RAM; GPU pod on a host-attached NVIDIA RTX 5090 via the device plugin). For each (backend, model, image) triple, 5 warm-up requests followed by 100 measured requests were issued *sequentially*, that is at concurrency one: exactly one request is in flight at any moment, so the measurement isolates per-request latency and excludes queuing effects. Latency was the client-side wall-clock HTTP round trip, reported as median and tail percentiles in line with established practice [Dean and Barroso, 2013]. The data set totals 5,040 requests with no failures.

5 Results

5.1 Classification performance

Table 2 reports test-set metrics. The three backbones cluster at 97.3–98.3% under the ImageNet regime; every pairwise difference between them has a confidence interval spanning zero, so they are statistically indistinguishable despite a tenfold range in parameter count. Victorio reaches 84.2%, a gap of 14.1 pp to the best backbone, far outside the interval. Hypothesis H1 is not supported.

Pretraining improves accuracy for all three backbones, by 6.6 pp (MobileNetV2), 2.9 pp (ResNet-50) and 1.7 pp (EfficientNet-B0). The direction is consistent, but only the MobileNetV2 difference is clearly outside sampling error; the ResNet-50 difference is marginal and the EfficientNet-B0 difference is within it. The Victorio pair, which differs in normalisation alone, separates by 0.4 pp and serves as a negative control.

Per-class results expose an asymmetry, visible in Figure 2. Under the ImageNet regime all three backbones classify Pneumonia perfectly (160/160, precision and recall 1.000), and all residual errors fall on the TB/Normal boundary. Pneumonia is also the only class drawn from a single source, which motivates Section 5.3. Tuberculosis recall differs little between active and obsolete disease for the backbones (for example 0.979 and 1.000 respectively for MobileNetV2 under the ImageNet regime), though the 26 obsolete test cases support only an indicative estimate.

5.2 Explainability and Agreement Index

Table 3 reports the Grad-CAM versus LIME Agreement Index at the 20% mask threshold. The rank ordering across model–regime pairs is preserved at 10%, 20% and 30% (Spearman $\rho \geq 0.976$, $p < 0.001$), so the index reflects a property of the attributions rather than of the threshold.

Agreement is *not* invariant to capacity. It varies from 0.132 to 0.243, and the ImageNet regime yields higher

Model	Regime	Accuracy [95% CI]	Macro F1	AUROC	TB recall
ResNet-50	ImageNet	0.983 [0.967, 0.992]	0.983	0.999	0.988
MobileNetV2	ImageNet	0.981 [0.965, 0.990]	0.981	0.999	0.981
EfficientNet-B0	ImageNet	0.973 [0.954, 0.984]	0.973	0.998	0.944
EfficientNet-B0	CXR	0.956 [0.934, 0.971]	0.956	0.994	0.925
ResNet-50	CXR	0.954 [0.932, 0.970]	0.954	0.988	0.969
MobileNetV2	CXR	0.915 [0.886, 0.936]	0.915	0.981	0.881
Victorio	ImageNet	0.842 [0.806, 0.872]	0.839	0.933	0.694
Victorio	CXR	0.838 [0.802, 0.868]	0.836	0.935	0.706

Table 2: Test-set performance (n = 480, 160 per class) with Wilson score intervals. Rows are ordered by accuracy; the ordering is not a ranking, as differences below roughly 3 pp are not resolvable at this sample size. Victorio has no ImageNet weights, so its two rows compare normalisation only.

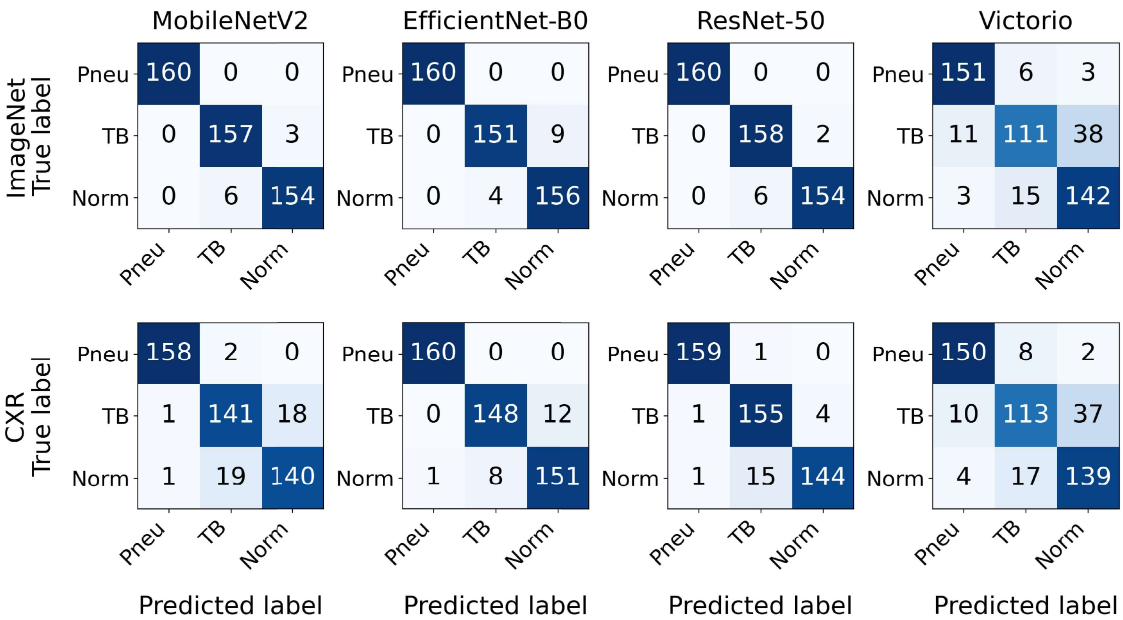

Figure 2: Confusion matrices on the held-out test set ($n = 480$). Top row: ImageNet regime; bottom row: CXR regime. Columns: MobileNetV2, EfficientNet-B0, ResNet-50, Victorio. Pneumonia is perfectly separated by all three backbones under the ImageNet regime, and residual errors concentrate on the TB/Normal boundary. Axis labels are abbreviated: Pneu (Pneumonia), Norm (Normal).

agreement than the CXR regime for every backbone. Victorio, whose two regimes differ only in normalisation, returns 0.143 in both, again acting as a negative control. Absolute values must be read against the mask areas: the Grad-CAM mask covers 20% of the image by construction while the LIME mask covers 55–85%, so the attainable maximum is the ratio of the two areas, approximately 0.29, rather than one. Agreement is therefore close to its ceiling for the pretrained backbones and materially below it for the compact network. By class, agreement is highest for Normal (0.199), then Pneumonia (0.187), then TB (0.155), consistent with focal tuberculosis lesions being a harder target for two methods to localise identically. Figure 3 shows both attributions for a representative case.

5.3 Source confound

Three measurements quantify the extent to which acquisition source, rather than pathology, is available to the models.

First, with the diagnosis held fixed, images sharing an acquisition source are more similar in raw pixels than images from different sources: within Normal, mean correlation 0.711 against 0.619 (t = 153.1); within TB, 0.604 against 0.574 ($t = 67.2$). Source leaves a signature independent of pathology.

Regime	MobileNetV2	EfficientNet-B0	ResNet-50	Victorio
ImageNet	0.216 ± 0.011	0.243 ± 0.011	0.199 ± 0.008	0.143 ± 0.014
CXR	0.178 ± 0.017	0.183 ± 0.011	0.132 ± 0.025	0.143 ± 0.013

Table 3: Grad-CAM versus LIME Agreement Index (IoU) at the 20% mask threshold, mean ± 95% confidence interval over correctly classified test images (n = 52–60 per cell). The attainable maximum is approximately 0.29 given the mask areas involved.

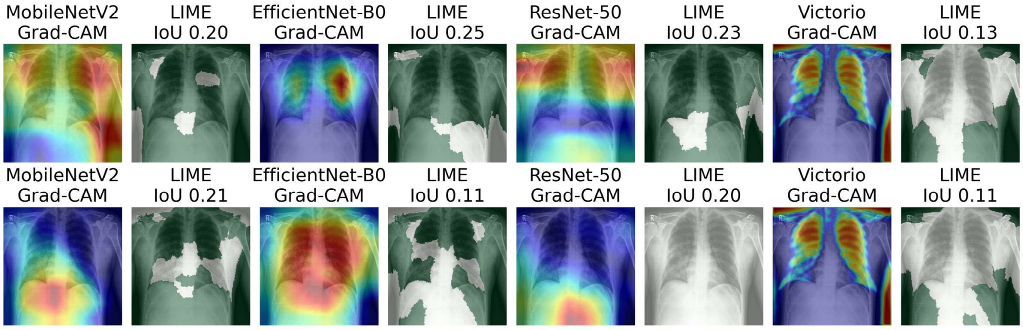

Figure 3: Grad-CAM and LIME on a representative Pneumonia test case. Top row: ImageNet regime; bottom row: CXR regime. Each pair of panels shows the Grad-CAM heatmap and the LIME positive-superpixel mask for the same case, with the Agreement Index for that case in the panel title.

Model	Regime	CPU pod		GPU pod	
		P50 (ms)	P95 (ms)	P50 (ms)	P95 (ms)
MobileNetV2	CXR	101	924	93	262
MobileNetV2	ImageNet	69	507	93	273
EfficientNet-B0	CXR	161	1010	110	300
EfficientNet-B0	ImageNet	188	1159	112	301
ResNet-50	CXR	517	1430	250	432
ResNet-50	ImageNet	398	1445	248	431
Victorio	CXR	19	113	48	204
Victorio	ImageNet	19	111	47	210

Table 4: Warm-state per-request latency at concurrency one (300 measured requests per row). Median (P50) and 95th percentile (P95) in milliseconds.

Second, a MobileNetV2 classifier trained to predict source rather than diagnosis separates CheXpert from the rest perfectly, but that is an upper bound rather than a measure of confounding, because CheXpert membership and the Pneumonia label partition the corpus identically. The informative test holds diagnosis fixed: within the tuberculosis class, TBX11K is separated from Montgomery, Shenzhen, DA and DB at 0.982 balanced accuracy and 0.999 AUROC against a 0.781 majority baseline. The equivalent test within Normal reaches 1.000 but rests on only 10 non-TBX11K test images.

Third, the checkpoints were evaluated where source is constant: TB against Normal, both from TBX11K, 275 test images. Performance is retained (ResNet-50 ImageNet 0.986, MobileNetV2 0.989, EfficientNet-B0 0.975, Victorio 0.836) and TB recall is essentially unchanged (0.984 against 0.988). This two-class problem has a 0.545 majority baseline and is not comparable with the three-class headline, but it establishes that TB/Normal discrimination does not depend on identifying the dataset of origin.

5.4 Deployment latency

Table 4 reports warm-state per-request latency for all eight pairs on both backends, aggregated over 100 measured requests per (backend, model, image) cell across three representative test images. Predicted classes were identical across CPU and GPU for every cell, confirming functional equivalence of the deployed models across devices.

The CPU-versus-GPU comparison is architecture-dependent. GPU halved the median latency of ResNet-50 (250 ms against 517 ms under the CXR regime, a 2.1× speed-up) and gave a smaller but consistent advantage on EfficientNet-B0 (110 ms against 161 ms, roughly 1.5×). For MobileNetV2 the two backends were within tens of milliseconds and effectively tied. Only for the compact custom CNN did CPU clearly win, at 19 ms against 48 ms, a 2.5× inversion. Tail behaviour also diverged: CPU P95 reached up to three times its own median (1,430 ms against 517 ms for ResNet-50 CXR), whereas GPU P95 stayed within roughly twice the median across all models.

6 Discussion

Capacity saturates, and pretraining helps. H1 is not supported: a 6,627-parameter network trails the backbones by 14 pp. The more useful observation is where the gap is not: MobileNetV2 at 2.2 M parameters matches ResNet-50 at 23.5 M to within sampling error, so capacity beyond roughly two million parameters buys nothing measurable on this task. Pretraining raises accuracy for every backbone, though only the MobileNetV2 difference is clearly resolvable at this sample size. Read alongside Section 5, the practical implication is that the choice worth making is pretrained-versus-scratch, not large-versus-small.

The source confound is real, and partly bounded. The corpus is not source-balanced, and Section 5.3 shows the consequence is not hypothetical: acquisition source is recoverable from raw pixels with the diagnosis held fixed, and a classifier recovers it at 0.982 balanced accuracy within the tuberculosis class. Performance nonetheless holds on a single-source subset, so TB/Normal discrimination reflects radiographic content rather than dataset identity. Pneumonia is the unresolved case: it is drawn entirely from CheXpert, no within-source control is available, and its perfect classification may in part reflect acquisition characteristics. Resolving this requires a pneumonia set from a second source, or domain-adversarial training [Ganin et al., 2016], and is beyond the scope of this study. The reported accuracies should be read as an upper bound on clinical performance.

Explainer agreement is not invariant to capacity. Agreement varies with both architecture and initialisation, and is consistently higher for pretrained models. That the Victorio pair returns an identical value under both regimes, while every backbone pair does not, indicates the variation tracks the representation rather than the measurement. Interpreted against the attainable ceiling of approximately 0.29, the pretrained backbones agree close to that limit while the compact network does not, which is consistent with a less spatially specific representation. We report this as an observation about two attribution methods, not as evidence that either localises pathology correctly; the sanity-check literature [Adebayo et al., 2018] cautions against the stronger reading.

Backend choice is architecture-dependent (H2). H2 is partially supported: the conventional expectation that CPU serving suits small models holds only at the lower bound of capacity. GPU per-request overhead (kernel launch, host-to-device transfer, stream synchronisation) is roughly constant, so it is amortised by the heavier forward passes of ResNet-50 and EfficientNet-B0 but dominates the compute of the compact network, inverting the ordering. For low-throughput single-image screening the guidance is split: pair compact backbones with CPU pods, retain GPU serving for ResNet-class architectures. The CPU pod also showed a heavy right tail, so GPU serving is preferable when worst-case rather than typical latency is the constraint.

Limitations. This is a pilot study on a single workstation, trained on one dataset combination. The 480-image test set limits resolution to roughly 3 pp, so the three backbones cannot be separated. All models share one hyperparameter configuration, which controls for tuning effort but disadvantages architectures whose optimum lies elsewhere; the from-scratch ResNet-50 run showed unstable validation behaviour, and each configuration was trained once, so no seed variance is reported. The tuberculosis class mixes active and healed disease. The benchmark ran at concurrency one on a single-node Minikube cluster; concurrent workloads would likely shift the CPU/GPU balance. Clinical translation would require validation under regulatory frameworks (for example EU MDR and GDPR).

7 Conclusion

Three findings stand out from this pilot. Capacity saturates early: MobileNetV2 at 2.2 M parameters is indistinguishable from ResNet-50 at 23.5 M, while the 6,627-parameter network trails by 14 pp. Grad-CAM and LIME agreement varies systematically with architecture and initialisation rather than being invariant to capacity, and is stable across mask thresholds. The deployment benchmark shows an architecture-dependent crossover, with GPU 2.1× faster than CPU on ResNet-50 and 2.5× slower on the compact network, so compact backbones should be paired with CPU pods while GPU serving remains right for heavier architectures. The corpus is not source-balanced, and the reported accuracies are an upper bound on clinical performance.

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